+86-13267237090, +92-3339533493
saeediqbalkhattak@gmail.com
orcid: 0000-0002-3176-4658

# Saeed Iqbal

Deep Learning Scientist

saeediqbal.github.io
linkedin.com/in/saeed-iqbal-khattak
Google Scholar: Cj7u9FkAAAAJ

## Research Interest

My research interests lie primarily in leveraging deep learning methodologies to integrate and analyze large-scale collections of multi-modal medical images, as well as advancing techniques in object detection and tracking. I began my **research career in 2009**, focusing on smartphone platform security, specifically for Android, with an emphasis on usage control and attestation. During my graduate studies, my focus shifted to **machine learning** and **deep learning** models, which I applied to support **medical image analysis** for tumor detection. I have hands-on experience working with radiology images and implementing advanced techniques, including **Machine Unlearning**, **Continual Learning**, **Federated Learning**, **Adaptive Self-Learning**, and **Anomaly Detection**, alongside a range of supervised and unsupervised learning methods. My Ph.D. research further solidified my expertise in medical image analysis using deep learning. Looking ahead, a key priority is to establish my own dedicated research laboratory, with support from the host university or organization, to drive further knowledge advancement and innovation in this critical field.

## PERSONAL INFORMATION

Born on August 10, 1984, in Karak, Khyber Pakhtunkhwa (KP) and formerly known as the North-West Frontier Province (NWFP), Pakistan. I possess over **20 years** of combined professional experience, including more than **17 years** in teaching and **3 years and 9 months** in industry roles. My post-PhD. experience extends for 3 years and 1 months. My research influence is demonstrated by **903 citations**, an **h-index of 16**, and an **i10-index of 21**.

## COMMITTEE MEMBERSHIP

I have actively contributed to university committees to shape academics, uphold quality, and drive excellence, with experience including:

1. **Program and Curriculum Development:** I have played a key role in designing and implementing new programs (Bachelor's in Data Science & AI, Master's in Data Science). For four years, I also served on the curriculum committee to align programs with industry needs.
2. **Quality and Development:** I have served on committees to maintain educational and research quality. I led a significant Outcome-Based Education (OBE) software development project for the Computer Science, Software Engineering, and Data Science programs within the Faculty of IT & Computer Science, UCP Lahore.
3. **Creative Works and Excellence Awards:** I have evaluated creative works and recognized academic excellence. My contributions include judging ICPC and other programming competitions at UCP, and serving as the Program Chair for the IEEE INMIC conference hosted at UCP, Lahore.
4. **Administrative Committees:** I led several critical academic committees for up to eight years, including the Exam Committee, Student Registration Committee, and Course Planning Committee for BSCS, BSSE, and BSDS programs.

## Awards & Grants

- ☆ Honored as a **Level-1 Researcher** in the year **2024, 2023, & 2022**
- ☆ Awarded Faculty wise **Best Research & Innovator Award** **2022, & 2023**
- ☆ Secured a prestigious grant from Samsung Innovation Campus (SIC) to offer cutting-edge Artificial Intelligence Training programs catering to undergraduates, graduates, and esteemed faculty members, **2021**.
- ☆ **Awarded** a grant from the Shenzhen Natural Science Foundation in Basic Research Fund, Shenzhen, China, for a project titled "SymPhyNeuMA: Symbolic-Physics Neural Material Adaptor for Visual Grounding of Industrial Anomaly Detection".

## Education

**Doctor of Philosophy (Computer Science)**, *University of Central Punjab*, *Pakistan* — AUG 2023
**Title:** *Adaptive Self-learning Systems for Medical Image Analyses using Hybrid-Dense Convolutional Neural Network*
**Supervisor:** *Dr. Adnan N. Qureshi*

**Master of Science (Computer Science)**, *National University of Computer & Emerging Sciences*, *Pakistan* — DEC 2012
**Title:** *Extending Java Pathfinder(JPF) with Property Classes for Verification of Android Permission Extension Framework*
**Supervisor:** *Dr. Shakir Ullah Shah*

**Bachelor of Information Technology (Honors)**, *Punjab University College of Information Technology*, *Pakistan* — OCT 2007
**Higher Secondary School Certificate**, *BISE, Lahore*, *Pakistan* — JUL 2003
**Secondary School Certificate**, *BISE, Peshawar*, *Pakistan* — JUN 2000

## Intellectual Property & Copyright (PATENT)

1. **Holistic Campus Security** optimized detection/tracking system using Deep Learning with commodity hardware (PTZ cameras) in real-time environment.

## Publication

52. **Iqbal, S.**, Zhong, X., Khan, M. A., Wu, Z., Almujally, N. A., Liu, W., Albalwy, F., & Hussain, A. (2026). Hierarchical federated learning with paillier encryption: synergistic approach for secure analytics of sensitive healthcare data, Expert Systems with Applications, 330, 132969, doi: 10.1016/j.eswa.2026.132969, **impact factor: 7.5**, Elsevier (2026)
51. **Iqbal, S.**, Zhong, X., Khan, M. A., Wu, Z., Almujally, N. A., Liu, W., Cambria, E., & Hussain, A. (2026). Responsible AI in Healthcare: Mitigating Hallucinations and Enhancing Multimodal Fusion-Based Reasoning in Medical Imaging, Information Fusion, 104483, doi: 10.1016/j.inffus.2026.104483, **impact factor: 15.5**, Elsevier (2026)
50. Ullah, M. S., Khan, M. A., **Iqbal, S.**, Alsenan, S., Alasiry, A., Marzougui, M., & Nam, Y. (2026). A Bayesian Backed DeepLabV3+ Customized Hybrid-Depth Framework for Brain Tumor Segmentation and Classification, IEEE Access, 14, 3658607, doi: 10.1109/ACCESS.2026.3658607, **impact factor: 3.4**, IEEE (2026)
49. Rahman, T. U., Li, G., Khan, A. D., Ali, F., **Iqbal, S.**, Kamal, S., & Ouyang, Z. (2026). Hybrid Adversarial Contrastive Framework for Robust Optical Free-Space Communication Under Atmospheric Turbulence and Domain Shift, Engineered Science, 41, 2290, doi: 10.30919/es2290, **impact factor: 6.6**, Engineered Science Publisher (2026)
48. Rahman, T. U., Li, G., Khan, A. D., Ali, F., **Iqbal, S.**, Nuaim, A. A., Ouyang, Z., Alruwaili, O., & Kamal, S. (2026). Dual-beam differential optical imaging for atmospheric turbulence mitigation in free-space propagation, Optics Express, 34(10), 17770–17793, doi: 10.1364/OE.519000, **impact factor: 3.3**, Optica Publishing Group (2026)
47. **Iqbal, S.**, Choudhry, I. A., Khan, M. A., & Ullah, A. (2026). FedTA++: A Temporal-Aware Framework for Mitigating Spatial-Temporal Catastrophic Forgetting in Federated Continual Learning, 2026 International Conference on Data Science, Machine Learning, and Intelligence (DataSciMI), 1-6, doi: 10.1109/11523959, IEEE (2026)
46. **Iqbal, S.**, Khan, M. A., Mustafa, G., Ayouni, S., Aljohani, A., & Hussain, A. (2026). FedCapD: Federated Class-Incremental Learning via Capsule Distillation and Diffusion Replay, Neurocomputing, 132111, doi: 10.2026/j.neucom.2026.123654, **impact factor: 6.5**, Elsevier (2026)
45. **Iqbal, S.**, Zhong, X., Khan, M. A., Wu, Z., Almujally, N. A., Liu, W., & Hussain, A. (2026). Causal Continual Unlearning with Disentangled Anomaly Representations for Private Industrial Vision. Information Processing & Management, 63(2), 104417, doi: 10.1016/j.ipm.2026.104417, **impact factor: 7.4**, Elsevier (2026)
44. **Iqbal, S.**, Zhong, X., Khan, M. A., Wu, Z., Almujally, N. A., Liu, W., & Hussain, A. (2026). TADynFed: Dynamic Modality-Adaptive Federated Learning with Tissue-Aware Disentanglement for Cross-Disease Analysis. Artificial Intelligence In Medicine, 63(2), 104417, doi: 10.1016/j.aiim.2026.104417, **impact factor: 6.2**, Elsevier (2026)
43. **Iqbal, S.**, Khan, M. A., Jamel, L., Qureshi, A. N., Choudhry, I. A., & Hussain, A. (2026). xMagNet: Dynamic Magnification-Aware Fusion with Uncertainty Quantification for Robust Breast Cancer Histopathology, Neurocomputing, 132111, doi: 10.1016/j.neucom.2026.132745, **impact factor: 6.5**, Elsevier (2026)
42. **Iqbal, S.**, Zhong, X., Khan, M. A., Wu, Z., Almujally, N. A., Liu, W., & Hussain, A. (2026). Core unlearning: A multi-modal gradient-efficient architecture for exact and approximate model rewriting. Information Processing & Management, 63(2), 104417, doi: 10.1016/j.ipm.2025.104417, **impact factor: 7.4**, Elsevier (2026)
41. **Iqbal, S.**, Zhong, X., Khan, M. A., Wu, Z., Almujally, N. A., Liu, W., Schuller, BW. & Hussain, A. (2025). Cross-modal invariant learning with latent diffusion for reliable medical diagnosis under dynamic shifts. Neurocomputing, 132111, doi: 10.1016/j.neucom.2025.132111, **impact factor: 6.5**, Elsevier (2025)
40. **Iqbal, S.**, Zhong, X., Khan, M.A., Wu, Z., AlHammadi, D.A., & Liu, W. (2025). Family-based Continual Learning for Multi-Domain Pattern Analysis in Federated Frameworks with GCN and ViT , Neural Networks, Elsevier, doi: 10.1016/j.neunet.2025.107920 **impact factor: 6.3**, Elsevier (2025)
39. **Iqbal, S.**, Zhong, X., Khan, M.A., Wu, Z., Almujally, N.A., Liu, W. & Hussain, A. (2025). FairBias: Mitigating Bias in Medical Image Diagnosis with Mixed Noise and Class Imbalance, Neurocomputing, Elsevier, doi: 10.1016/j.neucom.2025.130910 **impact factor: 6.5**, Elsevier (2025)
38. **Iqbal, S.**, Zhong, X., Khan, M.A., Wu, Z., Hussain, A., Alsenan, S. & Liu, W. (2025). Adaptive Fuzzy Convolution Networks for Uncertainty-Aware Image Analysis in Ambiguous Environments, Expert Systems with Applications, Elsevier, doi: 10.1016/eswa.2025.05.029 **impact factor: 7.5**, Elsevier (2025)
37. **Iqbal, S.**, Choudhry, I.A., Ullah, I., Kaur, K., Choi, B.J. & Hassan, M.M., (2025). Domain adaptive FL for edge-enabled privacy-preserving MRI analysis, Alexandria Engineering Journal, Elsevier, doi: 10.1016/j.aej.2025.05.007 **impact factor: 6.8**, Elsevier (2025)
36. **Iqbal, S.**, Zhong, X., Khan, M.A., Shabaz, M., Wu, Z., AlHammadi, D.A., Liu, W., Algamdi, S.A. & Li, Y., (2025). Transforming Healthcare Diagnostics With Tensorized Attention and Continual Learning on Multi-Modal Data, IEEE Transactions on Consumer Electronics, IEEE, doi: 10.1109/TCE.2025.3563986 **impact factor: 10.9**, IEEE (2025)
35. **Iqbal, S.**, Zhong, X., Khan, M. A., Wu, Z., AlHammadi, D. A., Liu, W., & Choudhry, I. A. (2025). Hierarchical Continual Learning for Domain-Knowledge Retention in Healthcare Federated Learning, IEEE Transactions on Consumer Electronics, IEEE, doi: 10.1109/TCE.2025.3563909 **impact factor: 10.9**, IEEE (2025)

34. **Iqbal, S.**, Zhong, X., Khan, M. A., Wu, Z., AlHammadi, D. A., Liu, W., & Choudhry, I. A. (2025). Continual and wisdom learning for federated learning: A comprehensive framework for robustness and debiasing. Information Processing & Management, 62(5), 104157, Elsevier. doi: 10.1016/j.ipm.2025.104157 **impact factor: 7.4**, Elsevier (2025)
33. **Iqbal, S.**, Zhong, X., Alhussein, M., Wu, Z. , Aurangzeb, K., Liu, W. & Zhang, Y. (2025). FusionGCNN: An IoT-Based Novel Spatiotemporal Graph Convolutional Network for ECG Arrhythmia Detection. IEEE Internet of Things Journal, IEEE, doi: 10.1109/JIOT.2025.3560344 **impact factor: 8.9**, IEEE (2025)
32. **Iqbal, S.**, Qureshi, A. N., Alabdultif, A., Khan, F. & Jhaveri, R.H. (2025), Federated Autoencoder Model for Secure Medical Image Analysis with Privacy Preservation and Assurance, IEEE Journal of Biomedical and Health Informatics, IEEE, doi: 10.1109/JBHI.2025.3546300 **impact factor: 7.7**, IEEE (2025)
31. **Iqbal, S.**, Qureshi, A. N., Alhussein, M., Aurangzeb, K., Mahmood, A., & Azzuhri, S. R. (2024). Dynamic SelectOut and voting-based federated learning for enhanced medical image analysis. Machine Learning: Science and Technology, IOP, (2025), doi:10.1088/2632-2153/ada0a6, **impact factor: 6.0**, IOP Science.
30. **Iqbal, S.**, Qureshi, A. N., Alhussein, M., Aurangzeb, K., Zubair, M. & Hussain, A. (2024), A Novel Reciprocal Domain Adaptation Neural Network for Enhanced Diagnosis of Chronic Kidney Disease, Expert Systems, Wiley, (2024), doi:10.1111/exsy.13750, **impact factor: 3.0**, Wiley.
29. Choudhry, I. A., **Iqbal, S.**, Alhussein, M., Aurangzeb, K., Qureshi, A. N., & Hussain, A. (2024), A Novel Interpretable Graph Convolutional Neural Network for Multimodal Brain Tumour Segmentation, Cognitive Computation, Springer, (2024), doi:10.1007/s12559-024-10387-w, **impact factor: 4.30**, Springer, (2024).
28. Alharbi, T., & **Iqbal, S.**, Novel hybrid data-driven models for enhanced renewable energy prediction, Frontiers in Energy Research, Frontiers, (2024), doi:10.3389/fenrg.2024.1416201, **impact factor: 2.6**, Frontiers.
27. Zubair, M., Owais, M., Mahmood, T., **Iqbal, S.**, Usman, S. M., & Hussain, I. (2024), Enhanced Gastric Cancer Classification and Quantification Interpretable Framework Using Digital Histopathology Images, Scientific Reports, Nature, (2024), doi:10.1038/s41598-024-73823-9, **impact factor: 3.8**, Nature.
26. Choudhry, I. A., **Iqbal, S.**, Alhussein, M., Qureshi, A. N., Aurangzeb, K., & Naqvi, R. A. (2024), Transforming Lung Disease Diagnosis with Transfer Learning Using Chest X-Ray Images on Cloud Computing, Expert Systems, Wiley, (2024), doi:10.1111/exsy.13750, **impact factor: 3.0**, Wiley.
25. **Iqbal, S.**, Qureshi, A. N., Alhussein, M., Aurangzeb, K., & Khan, F. Akbar, M. A. (2024), From Data to Diagnosis: Enhancing Radiology Reporting with Clinical Features Encoding and Cross-Modal Coherence, IEEE ACCESS, IEEE, (2024), doi:10.1109/ACCESS.2024.3449929, **impact factor: 3.4**, IEEE.
24. Choudhry, I. A., **Iqbal, S.**, Alhussein, M., Aurangzeb, K., Qureshi, A. N., & Anwar, M. S., & Khan, F. (2024). Privacy-preserving AI for early diagnosis of thoracic diseases using IoTs: A federated learning approach with multi-headed self-attention for facilitating cross-institutional study, Internet of Things, Elsevier doi:10.1016/j.iot.2024.101296, **impact factor: 6.0**, Vol. 27, Elsevier, 101296.
23. **Iqbal, S.**, Qureshi, A. N., Alhussein, M., Aurangzeb, K., Choudhry, I. A., & Anwar, M. S. (2024). Hybrid deep spatial and statistical feature fusion for accurate MRI brain tumor classification. Frontiers in Computational Neuroscience, doi:10.3389/fncom.2024.1423051, **impact factor: 2.1**, Vol. 18, Frontiers, 1423051.
22. **Iqbal, S.**, Qureshi, A. N., Aurangzeb, K., Alhussein, M., Wang, S., Anwar, M. S. & Khan, F. (2024). Hybrid Parallel Fuzzy CNN Paradigm: Unmasking Intricacies for Accurate Brain MRI Insights, IEEE TRANSACTIONS ON FUZZY SYSTEMS, doi: 10.1109/TFUZZ.2024.3372608, **impact factor: 11.9**, (2024) IEEE.
21. **Iqbal, S.**, Qureshi, A. N., Aurangzeb, K., Alhussein, M., Haider, S. I., & Rida, I. (2023). AMIAC: Adaptive Medical Image Analyses and Classification, a Robust Self-Learning Framework. Neural Computing and Applications (NCAA), Springer, (2023), doi: 10.1007/s00521-023-09209-1, **impact factor: 6.0**, Springer
20. **Iqbal, S.**, Choudhry, I. A., & Lodhi, A. M. (2023), Intelligent Buffer Management Policy in Post Disaster Network Using DTN, IEEE $25^{th}$ International Multi Topic Conference (INMIC), doi: 10.1109/INMIC60434.2023.10465769, IEEE.
19. **Iqbal, S.**, Qureshi, A. N., Alhussein, M., Aurangzeb, K., & Anwar, M.S. (2023). AD-CAM: Enhancing Interpretability of Convolutional Neural Networks with a Lightweight Framework - From Black Box to Glass Box, IEEE Journal of Biomedical and Health Informatics, IEEE, doi: 10.1109/jbhi.2023.3329231 **impact factor: 7.7**, IEEE .
18. **Iqbal, S.**, Qureshi, A. N., Alhussein, M., Mustafa, G., Aurangzeb, K., Javeed, K., & Naqvi, R. A. (2023). Privacy-Preserving Collaborative AI for Distributed Deep Learning with Cross-sectional Data, Multimedia Tools and Applications, Springer, (2023), doi:10.1007/s11042-023-17202-y **impact factor: 3.6**, Springer .
17. Choudhry, I. A., Qureshi, A. N., Aurangzeb, K., **Iqbal, S.**, & Alhussein, M. (2023). Hybrid Diagnostic Model for Improved COVID-19 Detection in Lung Radiographs Using Deep and Traditional Features. Biomimetics, Vol. 8(5), 406,doi:10.3390/biomimetics8050406, **impact factor: 4.5**, MDPI.
16. **Iqbal, S.**, Qureshi, A. N., Alhussein, M., Mustafa, Choudhry, I.A., G., Aurangzeb, K., & Khan, T. M. (2023). Fusion of Textural and Visual Information for Medical Image Modality Retrieval using Deep Learning-Based Feature Engineering, IEEE ACCESS, IEEE, (2023), doi:10.1109/ACCESS.2023.3310245, **impact factor: 3.9**, IEEE .
15. **Iqbal, S.**, Qureshi, A. N., Alhussein, M., Aurangzeb, K., & Kadry, S. (2023). A Novel Heteromorphous Convolutional Neural Network for Automated Assessment of Tumors in Colon and Lung Histopathology Images. Biomimetics, 8(4), 370, doi:10.3390/biomimetics8040370, **impact factor: 4.5**, MDPI.

14. **Iqbal, S.**, Qureshi, A. N., Aurangzeb, K., & Javeed, K. (2023). Privacy-Preserving Collaborative AI for Distributed Deep Learning with Cross-sectional Data, IEEE $5^{th}$ International Conference on Bio-engineering for Smart Technologies (BioSMART), Paris, France, doi:10.1109/BioSMART58455.2023.10162106, 07-09 June 2023, IEEE.
13. **Iqbal, S.**, Qureshi, A. N., Li, J., Arshad, I., & Mahmood, T. (2023). Dynamic Learning for Imbalance Data in Learning Chest X-Ray and CT Images. HELIYON, (2023), doi:10.1016/j.heliyon.2023.e16807, **impact factor: 3.776**, Cell Press & Science Direct, Elsevier.
12. **Iqbal, S.**, Qureshi, A. N., Li, J., & Mahmood, T. (2023). On the Analyses of Medical Images using Traditional Machine Learning Techniques and Convolutional Neural Networks, Springer Archives of Computational Methods in Engineering, doi:10.1007/s11831-023-09899-9, **impact factor:8.171**, Springer Nature.
11. **Iqbal, S.**, Qureshi, A. N., Ullah, A., Li, J., & Mahmood, T. (2022). Improving the Robustness and Quality of Biomedical CNN Models through Adaptive Hyperparameter Tuning. Applied Sciences, 12(22), 11870, doi:10.3390/app122211870, **impact factor: 2.838**, MDPI.
10. **Iqbal, S.**, & Qureshi, A. N. (2022). A heteromorphous deep CNN framework for Medical Image Segmentation using Local Binary Pattern, pp:63466-63480, doi:10.1109/ACCESS.2022.3183331, **impact factor: 3.367** IEEE Access.
9. **Iqbal, S.**, & Qureshi, A. N., Deep-Hist: Breast cancer diagnosis through histopathological images using convolution neural network, Journal of Intelligent & Fuzzy Systems, pp: 1347-1364, doi: 10.3233/JIFS-213158, **impact factor: 1.739**, IOS Press.
8. **Iqbal, S.**, Qureshi, A. N., & Mustafa, G. (2022). Hybridization of CNN with LBP for Classification of Melanoma Images, CMC-COMPUTERS MATERIALS & CONTINUA, vol. 71(3), pp: 4915-4939, doi:10.32604/cmc.2022.023178, **impact factor: 3.860**, Tech Science, Press.
7. Shaheen, M., Ahsan, A., & **Iqbal, S.** (2021)., Data Mining of Scientometrics for Classifying Science Journals. Intelligent Automation and Soft Computing, vol.28(3), pp: 873-885, doi:10.32604/iasc.2021.016622, **impact factor: 1.647**, Tech Science, Press.
6. **Iqbal, S.**, Qureshi, A. N., & Akter, M. (2019, September)., Using Local Binary Patterns and Convolutional Neural Networks for Melanoma Detection, In Proceedings of SAI Intelligent Systems Conference, London, United Kingdom, (pp. 782-789), doi:10.1007/978-3-030-29513-4_58, Springer, Cham.
5. **Iqbal, S.**, Qureshi, A. N., & Lodhi, A. M. (2018)., Content based video retrieval using convolutional neural network, In Proceedings of SAI Intelligent Systems Conference, London United Kingdom, (pp. 170-186), doi:10.1007/978-3-030-01054-6_12, Springer, Cham.
4. **Iqbal, S.**, Choudhry, I. A., & Shabbir, K. (2016), Verification of Android Permission Extension Framework using SPF and JPF, International Journal of Computer Science and Information Security, vol. 14(10), 340.
3. **Iqbal, S.**, and Muhammad Shaheen., A machine learning based method for optimal journal classification, In Internet Technology and Secured Transactions (ICITST), 2013 8th International Conference for, pp. 259-264, doi:10.1109/ICITST.2013.6750202, IEEE.
2. Shaheen, M., & **Iqbal, S**. (2013, December). Labeled clustering a unique method to label unsupervised classes. In 8th International Conference for Internet Technology and Secured Transactions (ICITST-2013) (pp. 210-214), doi:10.1109/ICITST.2013.6750193, IEEE.
1. **Iqbal, S.**, Shah, S. U., Nauman, M., & Amin, M. (2013)., Extending java pathfinder (JPF) with property classes for verification of android permission extension framework, In 2013 IEEE 3rd International Conference on System Engineering and Technology (pp. 15-20), doi: 10.1109/ICSEngT.2013.6650135, IEEE.

## Publication in Progress

- **Iqbal, S.**, Zhong, X., Alhussein, M., Wu, Z. , Aurangzeb, K., Liu, W. & Zhang, Y. (2024). Enhanced Industrial Anomaly Detection Through a Hybrid Fuzzy CNN and Vision Transformer Framework with Federated Learning. IEEE TRANSACTIONS ON FUZZY SYSTEMS , IEEE, (2024), **Under Review**.
- **Iqbal, S.**, Qureshi, A. N., Aurangzeb, K., Alhussein, M., Zhang, Y., Naqvi, R.A., and Choudhary, I.A. "Novel Hierarchical Cascaded Vision-Transformers based Federated Learning for Chest Medical Imaging Analysis", BMC Medical Informatics and Decision Making, Springer, (2024), **Major Revision**.

## Research Project

- **My Antenatal Monitoring Assistant (MAMA),** Maternal and fetal health care is a global challenge in developed and developing countries. Complications occurring during pregnancy can result in the mother's and infant's death. The significant reasons associated with such adverse outcomes are the lifestyle of the mother as well as limited knowledge regarding pregnancies.
- **Skin Cancer Detector,** developing software that can detect skin cancer from images using computer vision without the painful and time-consuming process of laboratories and reduce the cost of laboratory tests.
- **Capturing Human Action for Smart Vehicles,** developing a system that will alert the driver on such scenarios by tracking eye movements, head movements, and gestures within specific time frames.
- **Human Action Recognition System,** optimized detection/tracking system using Deep Learning with commodity hardware (PTZ cameras) in real-time environment.
- **Breast Cancer Detection,** working on R&D of breast cancer to recognize cancer from images using computer vision and deep learning model with the help of PyTorch (fb.com) and Tensor flow (google.com)

+86-13267237090, +92-3339533493
saeediqbalkhattak@gmail.com
orcid: 0000-0002-3176-4658

Saeed Iqbal
Deep Learning Scientist

saeediqbal.github.io
linkedin.com/in/saeed-iqbal-khattak
Google Scholar: Cj7u9FkAAAAJ

## Experience

**Research Associate Professor** **NOV 2024 - Present**
*College of Mechatronics and Control Engineering, Shenzhen University, Shenzhen, China*

* As a research associate professor specializing in Federated Learning and Continual Learning for Industrial Vision systems, I focus on managing models for industrial anomaly detection and vision systems. My work includes optimizing hyperparameter tuning and overseeing research associates. I coordinate the publication of results and propose new initiatives in fields such as biomedical image processing, Continual Learning, feature recognition, and health informatics. Additionally, I contribute to securing funding for future projects by engaging in collaborative research and developing project proposals. My involvement in medical image analysis is more limited but still significant.

**Director** **JUL 2023 - OCT 2024**
*Centre for Applied Data Analytics (CADA), FOIT&CS, UCP, Lahore*

* Lead interdisciplinary research group in health informatics, bioinformatics, and machine learning.
* Developed intelligent solutions supporting evidence-based healthcare.
* Guided and mentored undergraduate, graduate, and Ph.D. students.
* Innovated IT and communication methodologies to enhance healthcare quality.

**Researcher** **SEP 2022 - SEP 2023 (WFH)**
*Beijing Engineering Research Center for IoT Software and Systems, Beijing, China*

* I manage medical image analysis models, with a particular emphasis on efficient hyperparameter tuning. In addition, I oversee research associates, plan results publications, and suggest initiatives in pertinent areas such as biomedical signal processing, AI, medical image analysis, feature recognition, and health informatics. Obtaining financing for future attempts entails actively participating in joint research and crafting project proposals.

**Assistant Professor** **JUL 2013 - OCT 2024**
*University of Central Punjab,* *Lahore, Pakistan*

* **Samsung Innovation Campus** – course for undergraduate and graduate.
  - Taught the landscape of data science, and AI modeling with the foundation of mathematics, and deliver the basic concepts of probability and linear algebra using Python.

* Designed and taught the following undergraduate Computer Science courses:
  - Object Oriented Programming (4 CHrs) (FALL 2022)
  - Programming for Big Data (3 CHrs) (SPRING 2022, 2021 & 2020, FALL 2021, 2020 & 2019)
  - Introduction to Computing (4 CHrs)– (FALL 2021, 2020, 2019 & 2014, Spring 2015, 2014 and Summer 2014)
  - Web Information Retrieval (3 CHrs) (FALL 2018, Summer 2018& 2017)
  - Web Application Development (3 CHrs) – (FALL 2018, 2016 & 2015, Summer 2017, Spring 2017)
  - Introduction to DBMS (4 CHrs) - (Spring 2019, 2018 & 2016, Summer 2016 & 2013, Fall 2015)
  - Operating Systems (4 CHrs) - (FALL & Spring 2013)
  - Programming Fundamentals (4 CHrs) - (FALL 2013)

* Designed and taught the following graduate (MS. and Ph.D.) Computer Science and Data Science courses:
  - Tools and Techniques in Data Science (Spring 2023).
  - Research Methodology (FALL 2022).

My teaching philosophy focuses on fostering students' intrinsic motivation through active, problem-based learning that challenges them to apply modern technology to real-world issues. I have extensive experience in academic administration, including coordinating student and faculty affairs, managing course and exam logistics, and ensuring a smooth and supportive learning environment.

**Lecturer** **SEP 2009 - APR 2013**
*HPS College (Govt),* *Hangu, KP*

* Designed & taught Computer Science courses for B.Sc and F.Sc levels.
* Conducted weekly lectures for programming language and physics courses, explaining complex concepts. Secured $2^{nd}$ prize in the final evaluation.
* Graded assignments and exams, held office hours for student discussions. Managed 1-hour lectures and supervised 3-hour labs.

**Sr. Software Engineer** **JAN 2006 - SEP 2009**
*iSOFT Technology* *Lahore, Pakistan*

* Led and managed a project team of technical engineers (including designers and developers) to ensure timely and budget-friendly delivery of project solutions.
* Investigate, analyze, design, develop, and implement applications for day-to-day operations, focusing on technology improvements, upgrades, and modifications.

+86-13267237090, +92-3339533493
saeediqbalkhattak@gmail.com
orcid: 0000-0002-3176-4658

Saeed Iqbal
Deep Learning Scientist

saeediqbal.github.io
linkedin.com/in/saeed-iqbal-khattak
Google Scholar: Cj7u9FkAAAAJ

## Graduate Supervision

I supervised master's level students studying medical image analysis.

- Diabetic Foot Ulcer Segmentation Using Context-Aware Hybrid Approach – Laiqa Imran – L1S21MSCS0003
- Hybrid Approach of Feature Engineering and Convolutional Neural Network for Images Classification – Izaz Ahmad – SAP26873
- Context-based Detection of Diabetic Retinopathy using Deep Learning – Shahbaz Ali – L1F21MSCS00053
- Classifying Breast Cancer Using a Hybrid Approach – Ramsha Saeed – L1F21MSCS0004

## Advising Final Year Project

During my academic career, I have had the privilege of supervising a significant number of undergraduate students. I am proud to have guided over **80 Final Year Projects (FYPs)** in the fields of **Artificial Intelligence** and **Machine Learning**. These projects covered a diverse range of topics, including medical image analysis (such as skin cancer and lung disease detection), real-time object recognition and tracking, and innovative applications of AI for daily life, such as prescription readers and language recognizer. My role was to mentor students in the entire process, from conceptualization and problem-solving to the implementation and evaluation of their AI-based systems.

## Skills

| | |
|---|---|
| **Deep Learning and Computer Vision API:** | **Expert Level** : Tensor flow, PyTorch, Keras, Open CV.<br>**Intermediate Level** : Google BigQurey, Tika, Jackrabbit, Nutch (Lucene & Solr) |
| **Programming Language:** | **Expert Level**: Java, Python, C, C++, JavaScript, HTML, LaTeX, AJAX, PHP, SQL, MySQL, Web Services, CSS and XML (DTD, Schema, DOM) |
| **Version Control & Software Configuration Management:** | **Beginner Level**: Distributed Version Control System (DVCS), Mercurial/MQ, Git/StGit & VCS (RCS, CVS, SVN, SCCS) |
| **Software Verification**: | **Intermediate Level**: Java Pathfinder (JPF), Simple PRomela Interpreter (SPIN) & NuSMV. |
| **Cloud Computing Technology**: | **Beginner Level:** Spark/Hadoop/Mapreduce (HBase, NoSQL (MongoDB, Cassandra), Mahout (Parallel Clustering/Classification Techniques) |
| **Quantitative Research**: | Mathematical optimization, Mathematical Modeling & MySQL |

## Editorial Services

| | |
|---|---|
| **Editorial Member** | BMC Biomedical Engineering |
| **Program Committee** | Served on the Program Committee of the 7th International Conference on IT Convergence and Security (ICITCS 2017) – the leading conference on Trusted & Security technologies.<br>Served on the Program Committee of IEEE Symposium on Computer Applications & Industrial Electronics - the leading conference on computing industry technologies |
| **Journal and Conference Reviews** | Reviewed articles for several journals including IEEE Transactions on Medical Imaging, IEEE ACCESS, Springer Archives of Computational Methods in Engineering, Applied Intelligence, SAGE Digital Health, Tech Science Computers, Materials & Continua, IOS Press Journal of Intelligent & Fuzzy Systems (JIFS) and Elsevier Knowledge-Based Systems, Computer Biology and Medicine, Pattern Recognition, Information Processing & Management, Engineering Applications of Artificial Intelligence, Biomedical Signal Processing and Control, Computer Methods and Programs in Biomedicine, and Medical Image Analysis. |

## Certificates

**Big Data and Hadoop**, *Udemy, Inc.* DEC 2016
**Hadoop, MR, Hive and Spark**, *Udemy, Inc.* DEC 2016
**Introduction to Python**, *Udemy, Inc.* DEC 2016
**Introduction to the Biology of Cancer**, *Coursera, Inc.* NOV 2021

## References

**Dr. Adnan N. Qureshi**, *Birmingham Newman University*, *Birmingham, UK* dranq@yahoo.com
**Dr. Khursheed Aurangzeb**, *Department of Computer Science, College of Computer and Information Sciences, King Saud University*, *Riyadh, Saudi Arabia* kaurangzeb@ksu.edu.sa
**Dr. Mohammad Nauman**, *Department of Computer Science, Effat College of Engineering, Effat University*, *Jeddah, Kingdom of Saudi Arabia* recluze@gmail.com